Advanced Programming - Doubly Linked List

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1 Class Index 1

1.1 Class List . . . 1

2 File Index 3

2.1 File List . . . 3

3 Class Documentation 5

3.1 DoubleLinkedList Class Reference . . . 5

3.1.1 Constructor & Destructor Documentation . . . 5

3.1.1.1 DoubleLinkedList() [1/3] . . . 5

3.1.1.2 DoubleLinkedList() [2/3] . . . 6

3.1.1.3 DoubleLinkedList() [3/3] . . . 6

3.1.1.4 ~DoubleLinkedList() . . . 6

3.1.2 Member Function Documentation . . . 6

3.1.2.1 DLL_BackwardDisplay() . . . 6

3.1.2.2 DLL_Clear() . . . 6

3.1.2.3 DLL_ForwardDisplay() . . . 6

3.1.2.4 DLL_GetEnd() . . . 6

3.1.2.5 DLL_GetIDX() . . . 6

3.1.2.6 DLL_InsertBack() . . . 6

3.1.2.7 DLL_InsertFront() . . . 7

3.1.2.8 DLL_InsertIDX() . . . 7

3.1.2.9 DLL_Next() . . . 7

3.1.2.10 DLL_Prev() . . . 7

3.1.2.11 DLL_RemoveBack() . . . 7

3.1.2.12 DLL_RemoveFront() . . . 7

3.1.2.13 DLL_RemoveIDX() . . . 7

3.1.2.14 operator=() [1/2] . . . 7

3.1.2.15 operator=() [2/2] . . . 7

3.1.3 Friends And Related Symbol Documentation . . . 8

3.1.3.1 iterator_system . . . 8

3.2 factory_system Class Reference . . . 8

3.2.1 Member Function Documentation . . . 8

3.2.1.1 createFromArray() . . . 8

3.2.1.2 createSampleList() . . . 8

3.3 iterator_system Class Reference . . . 8

3.3.1 Member Function Documentation . . . 9

3.3.1.1 begin() . . . 9

3.3.1.2 hasNext() . . . 9

3.3.1.3 hasPrev() . . . 9

3.3.1.4 iteratebackward() . . . 9

3.3.1.5 iterateforward() . . . 9

3.3.1.6 next() . . . 9

3.3.1.7 prev() . . . 10
3.3.1.8 rbegin() . . . 10
3.4 SX Class Reference . . . 10
3.4.1 Constructor & Destructor Documentation . . . 10
3.4.1.1 SX() . . . 10
3.4.2 Member Data Documentation . . . 10
3.4.2.1 data . . . 10
3.4.2.2 next . . . 10
3.4.2.3 prev . . . 10

4 File Documentation **11**
4.1 factory_system.cpp File Reference . . . 11
4.2 factory_system.h File Reference . . . 11
4.3 factory_system.h . . . 11
4.4 iterator_system.cpp File Reference . . . 11
4.5 iterator_system.h File Reference . . . 12
4.6 iterator_system.h . . . 12
4.7 main.cpp File Reference . . . 12
4.7.1 Function Documentation . . . 12
4.7.1.1 main() . . . 12
4.8 main.h File Reference . . . 13
4.9 main.h . . . 13
4.10 twosidelist.cpp File Reference . . . 13
4.11 twosidelist.h File Reference . . . 13
4.12 twosidelist.h . . . 13

Index **15**

Chapter 1

Class Index

1.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

DoubleLinkedList 5
factory_system 8
iterator_system 8
SX 10

Chapter 2

File Index

2.1 File List

Here is a list of all files with brief descriptions:

factory_system.cpp 11
factory_system.h 11
iterator_system.cpp 11
iterator_system.h 12
main.cpp 12
main.h 13
twosidelist.cpp 13
twosidelist.h 13

Chapter 3

Class Documentation

3.1 DoubleLinkedList Class Reference

`#include <twosidelist.h>`

Public Member Functions

- DoubleLinkedList ()
- DoubleLinkedList (const DoubleLinkedList &)=delete
- DoubleLinkedList & operator= (const DoubleLinkedList &)=delete
- DoubleLinkedList (DoubleLinkedList &&other) noexcept
- DoubleLinkedList & operator= (DoubleLinkedList &&other) noexcept
- ~DoubleLinkedList ()
- void DLL_InsertFront (int o)
- int DLL_RemoveFront ()
- void DLL_InsertBack (int o)
- int DLL_RemoveBack ()
- void DLL_InsertIDX (int o, int inx)
- int DLL_RemoveIDX (int inx)
- int DLL_GetIDX (int inx)
- int DLL_GetEnd (int inx)
- void DLL_Clear ()
- void DLL_ForwardDisplay ()
- void DLL_BackwardDisplay ()
- void DLL_Next ()
- void DLL_Prev ()

Friends

- class iterator_system

3.1.1 Constructor & Destructor Documentation

3.1.1.1 DoubleLinkedList() [1/3]

`DoubleLinkedList::DoubleLinkedList () [inline]`

3.1.1.2 DoubleLinkedList() [2/3]

```
DoubleLinkedList::DoubleLinkedList (
            const DoubleLinkedList & )  [delete]
```

3.1.1.3 DoubleLinkedList() [3/3]

```
DoubleLinkedList::DoubleLinkedList (
            DoubleLinkedList && other)  [inline], [noexcept]
```

3.1.1.4 ~DoubleLinkedList()

```
DoubleLinkedList::~DoubleLinkedList ()  [inline]
```

3.1.2 Member Function Documentation

3.1.2.1 DLL_BackwardDisplay()

```
void DoubleLinkedList::DLL_BackwardDisplay ()
```

3.1.2.2 DLL_Clear()

```
void DoubleLinkedList::DLL_Clear ()
```

3.1.2.3 DLL_ForwardDisplay()

```
void DoubleLinkedList::DLL_ForwardDisplay ()
```

3.1.2.4 DLL_GetEnd()

```
int DoubleLinkedList::DLL_GetEnd (
            int inx)
```

3.1.2.5 DLL_GetIDX()

```
int DoubleLinkedList::DLL_GetIDX (
            int inx)
```

3.1.2.6 DLL_InsertBack()

```
void DoubleLinkedList::DLL_InsertBack (
            int o)
```

3.1.2.7 DLL_InsertFront()

```
void DoubleLinkedList::DLL_InsertFront (
            int o)
```

3.1.2.8 DLL_InsertIDX()

```
void DoubleLinkedList::DLL_InsertIDX (
            int o,
            int inx)
```

3.1.2.9 DLL_Next()

```
void DoubleLinkedList::DLL_Next ()
```

3.1.2.10 DLL_Prev()

```
void DoubleLinkedList::DLL_Prev ()
```

3.1.2.11 DLL_RemoveBack()

```
int DoubleLinkedList::DLL_RemoveBack ()
```

3.1.2.12 DLL_RemoveFront()

```
int DoubleLinkedList::DLL_RemoveFront ()
```

3.1.2.13 DLL_RemoveIDX()

```
int DoubleLinkedList::DLL_RemoveIDX (
            int inx)
```

3.1.2.14 operator=() [1/2]

```
DoubleLinkedList & DoubleLinkedList::operator= (
            const DoubleLinkedList & )  [delete]
```

3.1.2.15 operator=() [2/2]

```
DoubleLinkedList & DoubleLinkedList::operator= (
            DoubleLinkedList && other)  [inline], [noexcept]
```

3.1.3 Friends And Related Symbol Documentation

3.1.3.1 iterator_system

```
friend class iterator_system  [friend]
```

The documentation for this class was generated from the following files:

- twosidelist.h
- twosidelist.cpp

3.2 factory_system Class Reference

```
#include <factory_system.h>
```

Static Public Member Functions

- static DoubleLinkedList createSampleList ()
- static DoubleLinkedList createFromArray (const int *values, int count)

3.2.1 Member Function Documentation

3.2.1.1 createFromArray()

```
DoubleLinkedList factory_system::createFromArray (
            const int * values,
            int count)  [static]
```

3.2.1.2 createSampleList()

```
DoubleLinkedList factory_system::createSampleList ()  [static]
```

The documentation for this class was generated from the following files:

- factory_system.h
- factory_system.cpp

3.3 iterator_system Class Reference

```
#include <iterator_system.h>
```

Public Member Functions

- bool hasNext () const
- int next ()
- bool hasPrev () const
- int prev ()

Static Public Member Functions

- static void iterateforward (DoubleLinkedList &list)
- static void iteratebackward (DoubleLinkedList &list)
- static iterator_system begin (DoubleLinkedList &list)
- static iterator_system rbegin (DoubleLinkedList &list)

3.3.1 Member Function Documentation

3.3.1.1 begin()

```
iterator_system iterator_system::begin (
            DoubleLinkedList & list)  [static]
```

3.3.1.2 hasNext()

```
bool iterator_system::hasNext () const
```

3.3.1.3 hasPrev()

```
bool iterator_system::hasPrev () const
```

3.3.1.4 iteratebackward()

```
void iterator_system::iteratebackward (
            DoubleLinkedList & list)  [static]
```

3.3.1.5 iterateforward()

```
void iterator_system::iterateforward (
            DoubleLinkedList & list)  [static]
```

3.3.1.6 next()

```
int iterator_system::next ()
```

3.3.1.7 prev()

```
int iterator_system::prev ()
```

3.3.1.8 rbegin()

```
iterator_system iterator_system::rbegin (
            DoubleLinkedList & list)  [static]
```

The documentation for this class was generated from the following files:

- iterator_system.h
- iterator_system.cpp

3.4 SX Class Reference

```
#include <twosidelist.h>
```

Public Member Functions

- SX (int o)

Public Attributes

- int data
- SX * prev
- SX * next

3.4.1 Constructor & Destructor Documentation

3.4.1.1 SX()

```
SX::SX (
            int o)  [inline]
```

3.4.2 Member Data Documentation

3.4.2.1 data

```
int SX::data
```

< Detailed description after the member

3.4.2.2 next

```
SX* SX::next
```

3.4.2.3 prev

```
SX* SX::prev
```

The documentation for this class was generated from the following file:

- twosidelist.h

Chapter 4

File Documentation

4.1 factory_system.cpp File Reference

```
#include "factory_system.h"
#include <iostream>
```

4.2 factory_system.h File Reference

```
#include "twosidelist.h"
```

Classes

- class factory_system

4.3 factory_system.h

Go to the documentation of this file.

```
00001 #ifndef ADV_CODING_PROJECT_1_FACTORY_SYSTEM_H
00002 #define ADV_CODING_PROJECT_1_FACTORY_SYSTEM_H
00003
00004 #include "twosidelist.h"
00005
00006 class factory_system {
00007 public:
00008     static DoubleLinkedList createSampleList();
00009     static DoubleLinkedList createFromArray(const int* values, int count);
00010 };
00011
00012 #endif
```

4.4 iterator_system.cpp File Reference

```
#include "iterator_system.h"
#include "twosidelist.h"
#include <iostream>
#include <stdexcept>
```

4.5 iterator_system.h File Reference

Classes

- class iterator_system

4.6 iterator_system.h

Go to the documentation of this file.

```
00001 #ifndef ADV_CODING_PROJECT_1_ITERATOR_SYSTEM_H
00002 #define ADV_CODING_PROJECT_1_ITERATOR_SYSTEM_H
00003
00004 class DoubleLinkedList;
00005 class SX;
00006
00007 class iterator_system {
00008     SX* current;
00009     explicit iterator_system(SX* start);
00010
00011 public:
00012     bool hasNext() const;
00013     int next();
00014     bool hasPrev() const;
00015     int prev();
00016
00017     static void iterateforward(DoubleLinkedList &list);
00018     static void iteratebackward(DoubleLinkedList &list);
00019
00020     static iterator_system begin(DoubleLinkedList& list);
00021     static iterator_system rbegin(DoubleLinkedList& list);
00022 };
00023
00024 #endif
```

4.7 main.cpp File Reference

```
#include <iostream>
#include "factory_system.h"
#include "iterator_system.h"
#include "twosidelist.h"
```

Functions

- int main ()

4.7.1 Function Documentation

4.7.1.1 main()

```
int main ()
```

Program's main function. Here we use our factory method to create a sample list to test the project. With this, we test forward and backward iteration, removing from index, getting from various points, as well as getting next and previous. Then we do some operations on an array.

4.8 main.h File Reference

4.9 main.h

Go to the documentation of this file.

```
00001 #pragma once
```

4.10 twosidelist.cpp File Reference

```
#include <iostream>
#include <stdexcept>
#include "twosidelist.h"
```

4.11 twosidelist.h File Reference

```
#include <iostream>
```

Classes

- class SX
- class DoubleLinkedList

4.12 twosidelist.h

Go to the documentation of this file.

```
00001 #pragma once
00002 #include <iostream>
00003
00004 class SX {
00005 public:
00006     int data;
00007     SX* prev;
00008     SX* next;
00009     SX(int o) : data(o), prev(nullptr), next(nullptr) {}
00010 };
00011
00012 class DoubleLinkedList {
00013 private:
00014     SX* head;
00015     SX* tail;
00016     int size;
00017
00018 public:
00019     friend class iterator_system;
00020     DoubleLinkedList() : head(nullptr), tail(nullptr), size(0) {}
00021     DoubleLinkedList(const DoubleLinkedList&) = delete;
00022     DoubleLinkedList& operator=(const DoubleLinkedList&) = delete;
00023     DoubleLinkedList(DoubleLinkedList&& other) noexcept
00024         : head(other.head), tail(other.tail), size(other.size) {}
00025     DoubleLinkedList& operator=(DoubleLinkedList&& other) noexcept {
00026         if (this != &other) {
00027             DLL_Clear();
00028             head = other.head;
00029             tail = other.tail;
```

```
00030             size = other.size;
00031             other.head = nullptr;
00032             other.tail = nullptr;
00033             other.size = 0;
00034         }
00035         return *this;
00036     }
00037     ~DoubleLinkedList() { DLL_Clear(); }
00038 
00039     void DLL_InsertFront(int o);
00040     int DLL_RemoveFront();
00041     void DLL_InsertBack(int o);
00042     int DLL_RemoveBack();
00043     void DLL_InsertIDX(int o, int inx);
00044     int DLL_RemoveIDX(int inx);
00045     int DLL_GetIDX(int inx);
00046     int DLL_GetEnd(int inx);
00047     void DLL_Clear();
00048     void DLL_ForwardDisplay();
00049     void DLL_BackwardDisplay();
00050     void DLL_Next();
00051     void DLL_Prev();
00052 };
00053 
```

Index

~DoubleLinkedList
    DoubleLinkedList, 6

begin
    iterator_system, 9

createFromArray
    factory_system, 8
createSampleList
    factory_system, 8

data
    SX, 10
DLL_BackwardDisplay
    DoubleLinkedList, 6
DLL_Clear
    DoubleLinkedList, 6
DLL_ForwardDisplay
    DoubleLinkedList, 6
DLL_GetEnd
    DoubleLinkedList, 6
DLL_GetIDX
    DoubleLinkedList, 6
DLL_InsertBack
    DoubleLinkedList, 6
DLL_InsertFront
    DoubleLinkedList, 6
DLL_InsertIDX
    DoubleLinkedList, 7
DLL_Next
    DoubleLinkedList, 7
DLL_Prev
    DoubleLinkedList, 7
DLL_RemoveBack
    DoubleLinkedList, 7
DLL_RemoveFront
    DoubleLinkedList, 7
DLL_RemoveIDX
    DoubleLinkedList, 7
DoubleLinkedList, 5
    ~DoubleLinkedList, 6
    DLL_BackwardDisplay, 6
    DLL_Clear, 6
    DLL_ForwardDisplay, 6
    DLL_GetEnd, 6
    DLL_GetIDX, 6
    DLL_InsertBack, 6
    DLL_InsertFront, 6
    DLL_InsertIDX, 7
    DLL_Next, 7
    DLL_Prev, 7
    DLL_RemoveBack, 7
    DLL_RemoveFront, 7
    DLL_RemoveIDX, 7
    DoubleLinkedList, 5, 6
    iterator_system, 8
    operator=, 7

factory_system, 8
    createFromArray, 8
    createSampleList, 8
factory_system.cpp, 11
factory_system.h, 11

hasNext
    iterator_system, 9
hasPrev
    iterator_system, 9

iteratebackward
    iterator_system, 9
iterateforward
    iterator_system, 9
iterator_system, 8
    begin, 9
    DoubleLinkedList, 8
    hasNext, 9
    hasPrev, 9
    iteratebackward, 9
    iterateforward, 9
    next, 9
    prev, 9
    rbegin, 10
iterator_system.cpp, 11
iterator_system.h, 12

main
    main.cpp, 12
main.cpp, 12
    main, 12
main.h, 13

next
    iterator_system, 9
    SX, 10

operator=
    DoubleLinkedList, 7

prev
    iterator_system, 9

SX, 10

rbegin
iterator_system, 10

SX, 10
data, 10
next, 10
prev, 10
SX, 10

twosidelist.cpp, 13
twosidelist.h, 13